

Experiment 4: Sorting Sensor Alerts by Severity

Aim

To implement both **Normal Bubble Sort** and **Optimized Bubble Sort** in Python and compare the number of comparisons.

Problem Statement

A monitoring system receives sensor alerts with different severity levels. Sort the alerts using Bubble Sort and compare the number of comparisons made by the normal and optimized versions.

Algorithm

Step 1: Start.

Step 2: Read the number of alerts.

Step 3: Read the severity values.

Step 4: Apply Normal Bubble Sort and count comparisons.

Step 5: Apply Optimized Bubble Sort and count comparisons.

Step 6: Display the sorted alerts and comparison counts.

Step 7: Stop.

Source Code (Python)

main.py

```
1 # Compare Plain and Optimized Bubble Sort
2
3 n = int(input("Enter number of alerts: "))
4 arr = []
5
6 print("Enter alert levels:")
7 for i in range(n):
8     arr.append(int(input()))
9
10 plain = arr.copy()
11 opt = arr.copy()
12
13 plain_comp = 0
14 opt_comp = 0
15
16 # Plain Bubble Sort
17 for i in range(n - 1):
18     for j in range(n - i - 1):
19         plain_comp += 1
20         if plain[j] > plain[j + 1]:
21             plain[j], plain[j + 1] = plain[j + 1], plain[j]
22
23 # Optimized Bubble Sort
24 for i in range(n - 1):
25     swapped = False
26
27     for j in range(n - i - 1):
28         opt_comp += 1
29         if opt[j] > opt[j + 1]:
30             opt[j], opt[j + 1] = opt[j + 1], opt[j]
31             swapped = True
32
33     if not swapped:
34         break
35
36 print("Sorted Alerts:", opt)
37 print("Plain Comparisons:", plain_comp)
38 print("Optimized Comparisons:", opt_comp)
```

Output

Output

```
Enter number of alerts: 10
Enter alert levels:
2
1
3
2
1
4
3
2
5
1
Sorted Alerts: [1, 1, 1, 2, 2, 2, 3, 3, 4, 5]
Plain Comparisons: 45
Optimized Comparisons: 44
```

=== Code Execution Successful ===

Complexity Analysis

Aspect	Plain Bubble Sort	Optimized Bubble Sort
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Best Case	$O(n^2)$	$O(n)$
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Worst Case	$O(n^2)$	$O(n^2)$
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Space	$O(1)$	$O(1)$
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Result

Both Bubble Sort methods sorted the alerts correctly. The Optimized Bubble Sort performed fewer comparisons by stopping early when the list became sorted.